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DATA
SET

DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

	1.5	1	4	-1.5	5.5	3	5.5
	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	0.1	0.7	0.3	-0.1	0.5	0.2	0.5

- a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.
- b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

a. H_0 : 兩母體之間沒有顯著差異
 H_a : 有顯著差異

$$T^+ = 1.5 + 1 + 4 + 5.5 + 3 + 5.5 = 26.5$$

$$T^- = 1.5$$

$$T = \min(T^+, T^-) = 1.5$$

$$\text{Critical Value } (n=7, \alpha=0.05 \text{ 雙尾}) = 2$$

$$T = 1.5 \leq 2 \Rightarrow \text{Reject } H_0 \Rightarrow \text{兩母體之間存在顯著差異}$$

$$(b) \text{ p-value} = P(X \leq 1) + P(X \geq 6) = \left(\binom{7}{0} + \binom{7}{1} + \binom{7}{6} + \binom{7}{7} \right) \times \left(\frac{1}{2} \right)^7 = 0.125$$

$\therefore \text{p-value} > 0.05 \Rightarrow \text{Non Reject } H_0$, 不同於 (a)

因 sign test 僅利用差異的正負方向, 而未考慮差異的幅度

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B		
1	3	7	-4	-6
2	14	12	2	3.5
3	7	9	-2	-3.5
4	10	15	-5	-1
5	9	12	-3	-5
6	6	6	0	
7	13	12	1	1.5
8	6	5	1	1.5
9	7	13	-6	-8

- a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
- b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

a. H_0 : 機器A、B的月故障率沒有顯著差異
 H_a : 有

$$T^+ = 1.5 + 1.5 + 3.5 = 6.5, \quad T^- = 3.5 + 5 + 6 + 1 + 8 = 29.5$$

$$T = \min(6.5, 29.5) = 6.5$$

$$\text{Critical Value } (n=8, \alpha=0.05 \text{ 雙尾}) = 4$$

$$T = 6.5 > 4 \Rightarrow \text{Non-Reject } H_0$$

b. 產量與工作負荷的月變化、季節及環境因素、原料差異
 是否剛保養完

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

- What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
- Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
- Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

a.

1. paired t-test, Sign Test
Wilcoxon Signed-Rank Test

2. 配對差值須來自一呈常態分布的母體

3. 樣本數過少，不能完美符合常態分布

b.

H_0 : 兩母體沒有顯著差異

H_a : 有

Sign test: $+$: 20, $-$: 0, $n = 20$

$X = 0$

$p\text{-value} = 2 \times P(X \leq 0) = 2 \times 0.5^{20} < 0.05$
 $\Rightarrow \text{Reject } H_0$

Wilcoxon Signed-Rank Test:

$T^+ = 210, T^- = 0 \Rightarrow T = 0$

Critical Value = 52

$T = 0 \leq 52 \Rightarrow \text{Reject } H_0$

c. (b) 的兩個結論相同，因在該極端案例中 With Imagery 的分數皆高於 Without Imagery $\Rightarrow$ 檢定皆能得出極度顯著的結果

Applying the Basics

DATA
SET

DS1518

6. **Legos®** The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)				
.5	1.0	1.5	2.0	
8	15	9	16	4
2	4	2	4	2
14	20	7	10	6
7	15	7	10	7
9	16	5	5	7
16	5	5	7	10
12	19	8	13	8
13	8	13	5	5
10	18	9	16	6
7	15	7	10	4
2	4	2	4	2

Use the Kruskal–Wallis H -Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

H_0 : 四種不同訓練時間組別的積木組裝時間分布沒有差異
 H_a : 至少有一組分佈與其它組不同

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1) = \frac{12}{420} \left[\frac{1296}{5} + \frac{2600}{5} + \frac{1600}{5} + \frac{576}{5} \right] - 63$$
$$= 12.27$$

$$df = k - 1 = 4 - 1 = 3, \alpha = 0.01, \chi^2_{3, 0.01} = 11.345$$

$12.27 > 11.345 \Rightarrow \text{Reject } H_0$, 四種不同訓練時間的幼稚園兒童, 其組裝積木的時間分布有顯著差異

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10-19	20-39	40-59	60-69
	29	24	37	28
	33	27	25	29
	26	33	22	34
	27	31	33	36
	39	21	28	21
	35	28	26	20
	33	24	30	25
	29	34	34	24
	36	21	27	33
	22	32	33	32
Total	309	275	295	282

241.5 168 216.5 188

- Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal-Wallis H -test with $\alpha = .01$.
- Find the approximate p -value for the test in part a.
- Since the F -test in Chapter 11 and the H -test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F -test produced the test statistic $F = 0.87$ with p -value = 0.468, compare this p -value with the p -value in part b and explain the implications of the comparison.

a. H_0 : 四組母體分布相同

H_a : 四組母體分布不完全相同

$$H = \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(N+1)$$

$$= \frac{12}{40(41)} \left[\frac{241.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right]$$

$$- 3 \times 41 = 2.632$$

$$df = 4 - 1 = 3$$

$$\chi^2_{3, 0.01} = 11.345$$

$$H = 2.632 < 11.345 \Rightarrow \text{Non-Reject } H_0$$

$$b. \chi^2_{3, 0.1} = 6.251$$

$$\Rightarrow p\text{-value} > 0.1$$

$$p\text{-value} \approx 0.462$$

c. ANOVA Test : $p\text{-value} = 0.468$

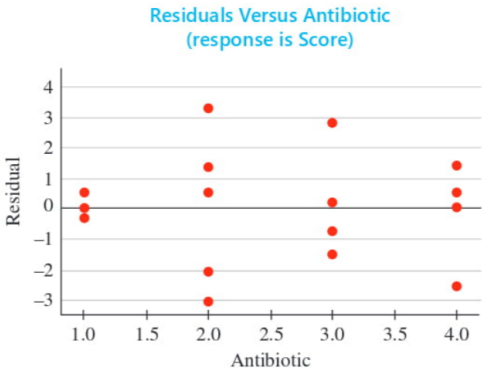
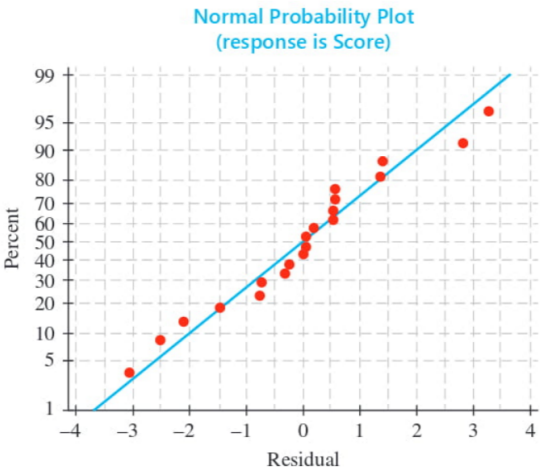
$H\text{-test} : p\text{-value} = 0.462 \Rightarrow$ 皆 reject H_0

Implication: 兩者結果相同, 結論 robustness

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.⁷ The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

Child	Antibiotic			
	1	2	3	4
1	4.8	2.2	6.8	6.2
2	8.1	9.2	6.6	9.6
3	5.0	2.6	3.6	6.5
4	7.9	9.4	5.3	8.5
5	3.9	7.4	2.1	2.0

- a. What design is used in collecting these data?
- b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?



- c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

a. Randomized Block Design

b. 常態性假設大致滿足
變異數同質性不符合

c. H_0 : 四種母體分布相同

H_a : 母體分布不完全相同

$$Fr = \frac{12}{5 \times 4 \times 5} (12^2 + 13^2 + 10^2 + 15^2) - 3 \times 5 \times 5$$
$$= 1.56$$

$$1.56 < \chi^2_{3, 0.05} = 7.815$$

$\Rightarrow$ Non-Reject H_0 .

11. Yield of Wheat Exercise 9 (Chapter 11

Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

Variety	Location					
	1	2	3	4	5	6
A	35.3 ₄	31.0 ₂	32.7 ₃	36.8 ₃	37.2 ₃	33.1 ₃
B	30.7 ₁	32.2 ₃	31.4 ₂	31.7 ₂	35.0 ₂	32.7 ₂
C	38.2 ₅	33.4 ₄	33.6 ₄	37.1 ₄	37.3 ₄	38.2 ₅
D	34.9 ₃	36.1 ₅	35.2 ₅	38.3 ₅	40.2 ₅	36.0 ₄
E	32.4 ₂	28.9 ₁	29.2 ₁	30.7 ₁	33.9 ₁	32.1 ₁

- a. Use the Friedman F_r -test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.
- b. When the analysis of variance was performed in Chapter 11, the ANOVA F -test produced the test statistic $F = 18.61$ with p -value = .000. How do these results compare with results of the Friedman F_r -test in part a? Explain.

$$R_A = 18$$

$$R_B = 12$$

$$R_C = 26$$

$$R_D = 21$$

$$R_E = 1$$

a. H_0 : 五種不同小麥的機率分佈是相同的
 H_a : 不完全相同

$$F_r = \left[\frac{12}{b k (k+1)} \sum_{i=1}^k R_i^2 \right] - 3b(k+1) = \left[\frac{12}{6 \times 5 \times (5+1)} (18^2 + 12^2 + 26^2 + 21^2 + 1^2) \right] - 3 \times 6 \times (5+1) = 20.133$$

$$20.133 > \chi^2_{4, 0.05} = 9.488 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 五種小麥品種的產量有顯著差異

b. p -value = 0.000 < 0.05 $\Rightarrow$ Reject H_0 , 結論與 (a) 一致